Mark schemes

Q1.

(a) $7 + 4x - 2x^2 = 9 - 2(x - 1)^2$

M1A1

2

(b) Put $u = \sqrt{2}(x - 1)$

allow $u = k(x - 1)$ any k

M1

$du = \sqrt{2}\,dx$

A1F

$I = \frac{1}{\sqrt{2}}\int\frac{du}{\sqrt{9-u^2}}$

ft error in (a); must have u^2 only, ie $\frac{1}{\sqrt{2}}$ outside integrand

A1F

$= \frac{1}{\sqrt{2}}\sin^{-1}\frac{u}{3}$

for $\sin^{-1}\frac{u}{p}$

A1

Change limits or replace u

provided $\sin^{-1}$

M1

$= \frac{\pi}{4\sqrt{2}}$ or $\frac{\pi\sqrt{2}}{8}$

CAO

A1

6

Alternative − if integration is attempted without substitution:
$\sin^{-1}$

(M1)

$\frac{1}{\sqrt{2}}$

(A1F)

$(x-1)$

(A1)

$\frac{\sqrt{2}}{3}$

(A1F)

Substitution of limits

(m1)

$\frac{\pi}{4\sqrt{2}}$

CAO

(A1) (6) [8]

Q2.

(a) $t = \tan\vartheta$ $\quad \mathrm{d}t = \sec 2\vartheta \, \mathrm{d}\vartheta$

OE

B1

$$I = \int \frac{\mathrm{d}t}{(9\cos^2\theta + \sin^2\theta)\sec^2\theta}$$

OE

M1

$$= \int \frac{\mathrm{d}t}{t^2+9}$$

AG

A1 3

(b) $$I = \left[\frac{1}{3}\tan^{-1}\frac{t}{3}\right]_0^{\sqrt{3}}$$

M1 for tan

M1

$$\frac{1}{3}\tan^{-1}\frac{\sqrt{3}}{3} \text{ or } \frac{1}{3}\tan^{-1}\frac{1}{\sqrt{3}}$$

A1

$= \frac{\pi}{18}$

AG

A1

3

[6]

Examiner reports

Q1.

It was surprising to see a number of candidates unable to do the algebra of part (a) correctly, and this did mean that it was not possible to score full marks on part (b). If a substitution was used in part (b), it was often $u = x - 1$, rather than $u = \sqrt{2}(x - 1)$, which meant that candidates lost a $\sqrt{2}$ in their solution. However there were a good number of completely correct solutions, many of which came from the direct integration of the integrand with no substitution.

Q2.

In part (a), whilst $\frac{dt}{d\theta}$ was expressed correctly, the manipulation required to obtain the integral in terms of t was frequently faulty. Also in part (b), whilst many candidates were able to write down the correct definite integral $\frac{1}{3}\tan^{-1}\frac{t}{3}$, full marks were not awarded unless it was clear how the answer $\frac{\pi}{18}$ was arrived at, as this answer was given in the question.

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